Eliminative Information in a Generalized Monty Hall Problem III

Heterogeneous Rewards and Strategic Initial Choice

Abstract

Parts I and II studied identical rewards. This note introduces heterogeneous nonnegative rewards. There are two distinct cases. If reward labels are exchangeable, then heterogeneity is mostly harmless: under uniform switching, the expected reward after t reveals depends only on total reward mass $V = \sum_i v_i$. If door labels carry different prior reward distributions, the problem changes qualitatively. The identity of Monty's zero-reward reveals matters, switching rules diverge, and the initial door can become a strategic sacrifice: choosing a low-prior door can preserve high-prior doors for later switching.

1 Two Kinds of Heterogeneity

Heterogeneous rewards can mean two different things.

Exchangeable rewards. There is a multiset

$$\{v_1, \ldots, v_m, 0, \ldots, 0\},$$

randomly permuted among K doors. Door labels carry no information.

Door-specific priors. Each door j has its own prior distribution for a nonnegative reward X_j. The player knows these priors but not the realized rewards. Monty observes the realized rewards and may reveal only zero-reward doors.

The first model gives a clean theorem. The second model is where the new decision theory appears.

2 Exchangeable Reward Multisets

Let positive rewards $v_1, \ldots, v_m$ be randomly placed among K doors, along with $K - m$ zeros. Let

$$V = \sum_{i=1}^{m} v_i.$$

The player chooses an initial door uniformly. Monty reveals t zero-reward doors, never opening the initial door. The player then switches uniformly among the remaining non-initial closed doors.

Theorem 1 (Collapse to total reward mass)**.** *Under exchangeable placement and uniform switching,*

$$\mathbb{E}[\textit{stay reward}] = \frac{V}{K}$$

and

$$\mathbb{E}[\textit{switch reward after } t \textit{ reveals}] = \frac{V(K-1)}{K(K-1-t)}.$$

Proof. The expected reward of the initial door is V/K by exchangeability. The total expected reward mass outside the initial door is $V(K-1)/K$. Monty reveals only zero-reward doors, so this reward mass is not changed by the reveals. After t reveals it is spread symmetrically over $K-1-t$ switch options. Uniform switching therefore has expected reward

$$\frac{1}{K-1-t} \cdot \frac{V(K-1)}{K} = \frac{V(K-1)}{K(K-1-t)}.$$

□

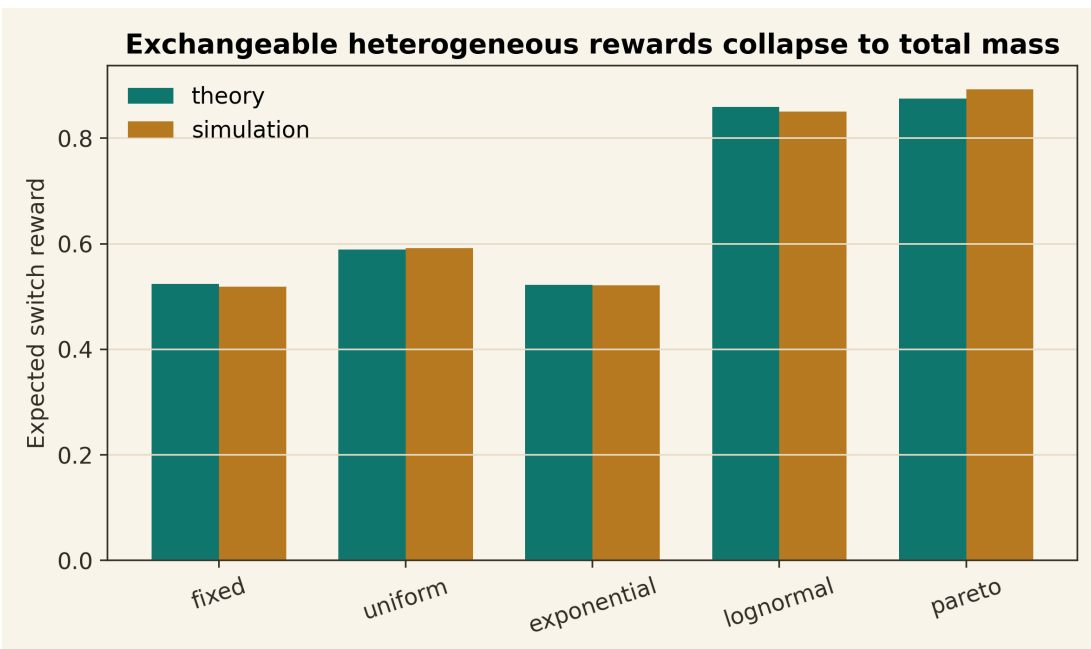

Figure 1: For exchangeable reward multisets, simulations match the total-mass formula across several reward distributions.

Remark 1. *If all positive rewards are unit rewards, then $V = m$ and this reduces to the Part I formula. Heterogeneity alone does not complicate the model. Labeled asymmetry does.*

3 Door-Specific Bernoulli-Value Priors

For a first labeled model, let each door have

$$X_j = \begin{cases} 0, & \text{with probability } q_j, \\ v_j, & \text{with probability } 1 - q_j. \end{cases}$$

The prior mean is

$$\mu_j = (1 - q_j)v_j.$$

This model has a real atom at zero, so Monty has legal zero-reward doors to reveal. It also makes door labels meaningful.

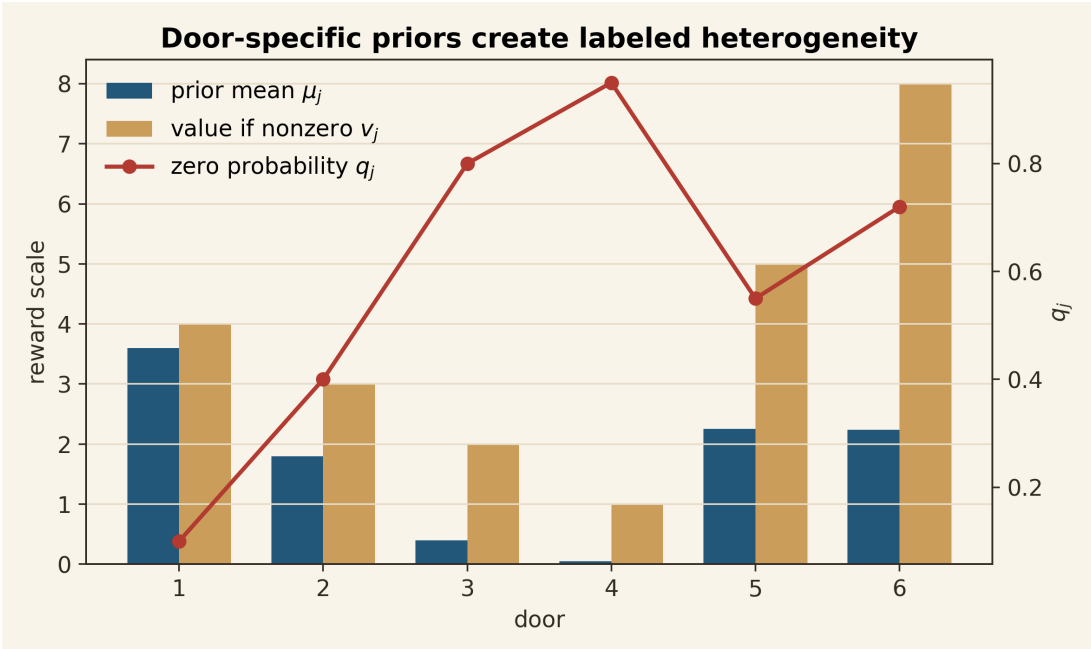

Figure 2: A labeled prior landscape: doors differ in zero probability, positive reward size, and prior mean.

4 Switching Rules Separate

With exchangeable doors, uniform switching is natural. With labeled priors, several switch rules differ:

- stay with the initial door;
- switch uniformly among available doors;
- switch to the available door with largest prior mean μ_j;
- switch to the available door with largest realized reward, an oracle upper bound.

5 Strategic Initial Choice

The initial choice is irrelevant in the symmetric model. It is not irrelevant with labeled priors. If the player expects to switch, choosing a high-mean door initially can be wasteful because that door is removed from the switch set. Choosing a low-mean door can preserve high-mean doors for later.

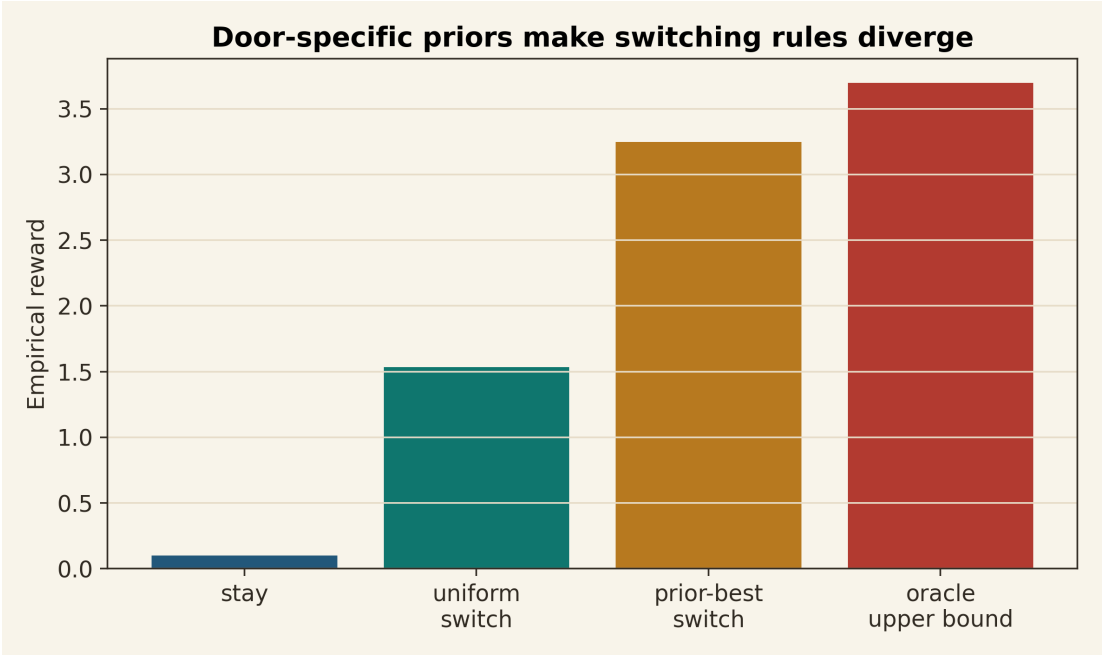

Figure 3: With door-specific priors, prior-best switching can substantially outperform uniform switching. The oracle bar is an infeasible upper bound.

Proposition 1 (Sacrifice choices can be valuable)**.** *There exist door-specific priors for which choosing the lowest-mean door as the initial door gives a larger expected prior-best switch reward than choosing the highest-mean door.*

Proof. The claim is constructive. Consider four doors with Bernoulli-value priors

j	1	2	3	4
q_j	0.10	0.40	0.80	0.95
v_j	4.00	3.00	2.00	1.00

so that the prior means are $3.6, 1.8, 0.4, 0.05$. If the player chooses door 1 initially, the best switch candidates exclude the highest-mean door. If the player chooses door 4 initially, doors 1 and 2 remain available for switching. Direct enumeration or simulation under uniform zero-revealing Monty shows that the latter policy has the larger prior-best switch reward. Figure 4 displays the comparison. □

6 Monty's Policy Now Matters

In the symmetric model, Monty's choice among legal zero doors is irrelevant. With labeled priors, it matters. Revealing a zero door with high prior mean is more informative than revealing a zero door that was already expected to be empty.

7 Concrete Numerical Examples

The simulations above are meant as illustrations, but several numerical examples are especially revealing.

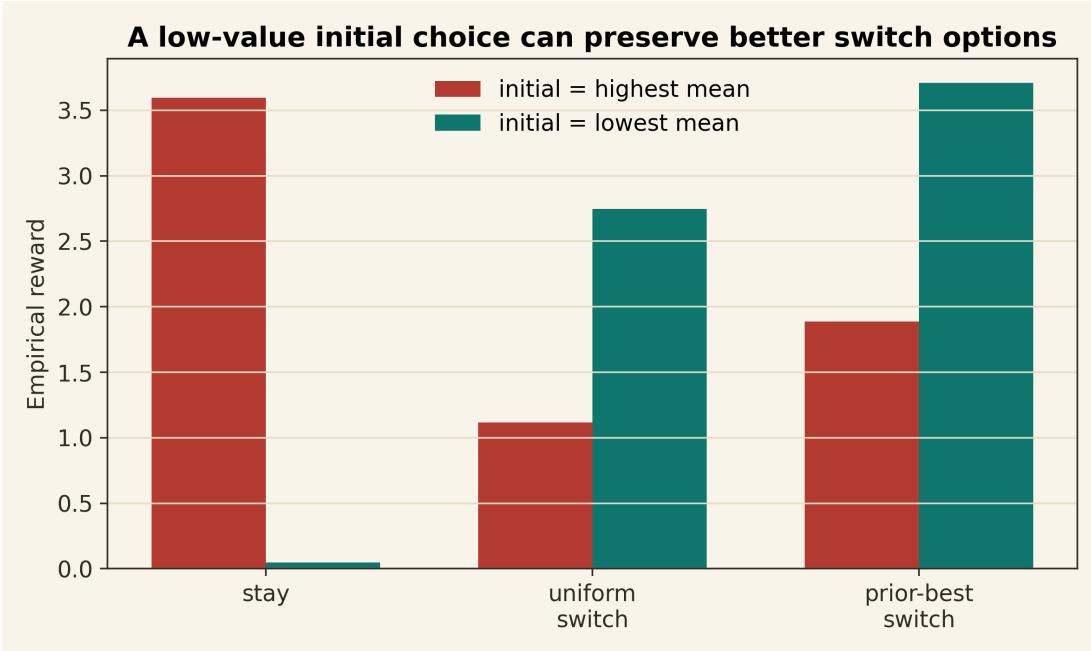

Figure 4: A low-value initial choice can preserve better switch options.

Exchangeable rewards remain scalar. With $K = 12$, $m = 4$, $t = 4$, and Pareto-distributed positive rewards, a run with 50,000 trials gave mean total reward

$$\bar{V} \approx 6.685.$$

The empirical and theoretical values were:

strategy	empirical reward	theory
stay	0.5549	0.5571
uniform switch	0.8774	0.8754

Even with heavy-tailed positive rewards, the exchangeable model behaves just as the theorem predicts. The individual reward distribution is not the main source of complexity; labeled information is.

Door-specific priors create large strategy gaps. In a door-specific Bernoulli-value model with $K = 10$, $t = 2$, random priors, and the lowest-mean door chosen initially, one run produced:

strategy	empirical reward
stay	0.073
uniform switch	0.585
prior-best switch	1.373
oracle switch	1.662

The prior-best strategy captures most of the oracle upper bound, while uniform switching leaves substantial value on the table. This is the practical meaning of door-specific priors: once doors have different expected values, "switch" is no longer a single strategy.

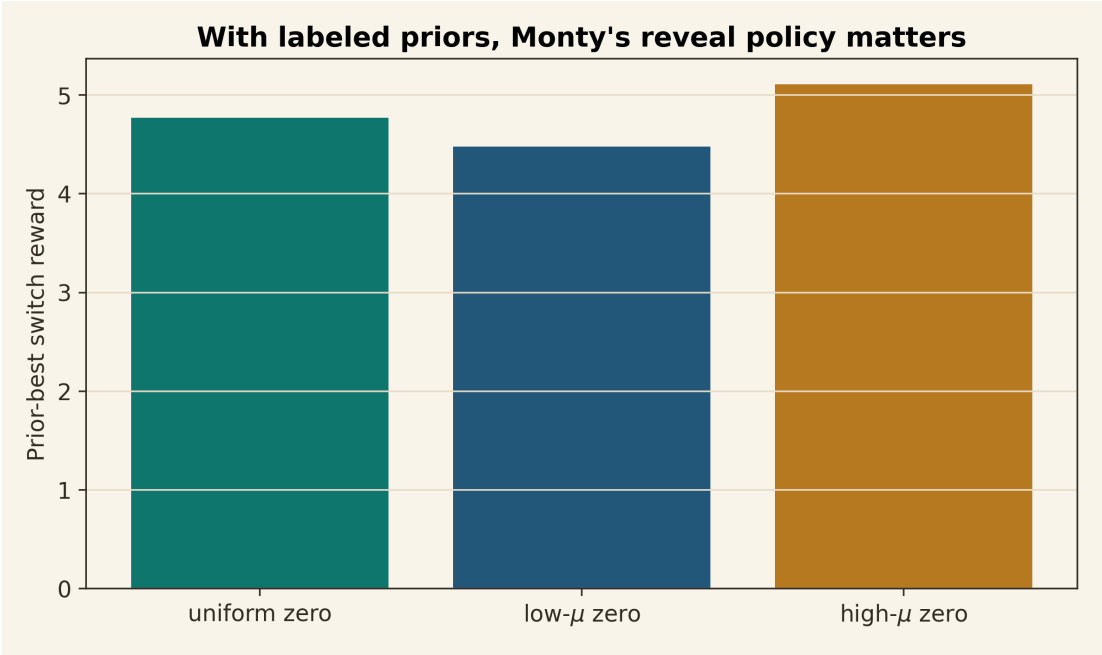

Figure 5: Different legal reveal policies can change the value of prior-best switching. This is the first point where host behavior becomes genuinely strategic.

The sacrifice effect can be dramatic. A search over four-door Bernoulli-value prior instances found the following door priors:

j	q_j	v_j	$\mu_j = (1-q_j)v_j$
1	0.256	2.745	2.042
2	0.642	2.390	0.855
3	0.329	28.833	19.342
4	0.230	0.450	0.347

Door 3 is overwhelmingly the best door by prior mean. If the player chooses the highest-mean door initially, then that door is unavailable as a switch target. If the player instead chooses the lowest-mean door initially, the high-value door remains available for switching.

The resulting empirical rewards were:

strategy	initial = highest mean	initial = lowest mean
stay	19.220	0.344
uniform switch	1.439	10.028
prior-best switch	2.245	19.729
oracle switch	2.328	20.023

This is the clearest numerical expression of the new phenomenon. In the classical problem the initial choice is a guess. Here it can be an instrument. If the plan is to switch, the player may rationally choose a bad initial door in order to preserve the best doors for the switch set.

8 Conclusion

Heterogeneous rewards split the theory into two regimes. Under exchangeable placement and uniform switching, the problem remains scalar: only total reward mass matters. Under door-specific priors, the problem becomes a labeled decision problem. The identity of revealed zero doors matters, switching rules separate, Monty's policy matters, and the initial choice can rationally be a sacrifice. This is the natural starting point for posterior updating and fully strategic host models.